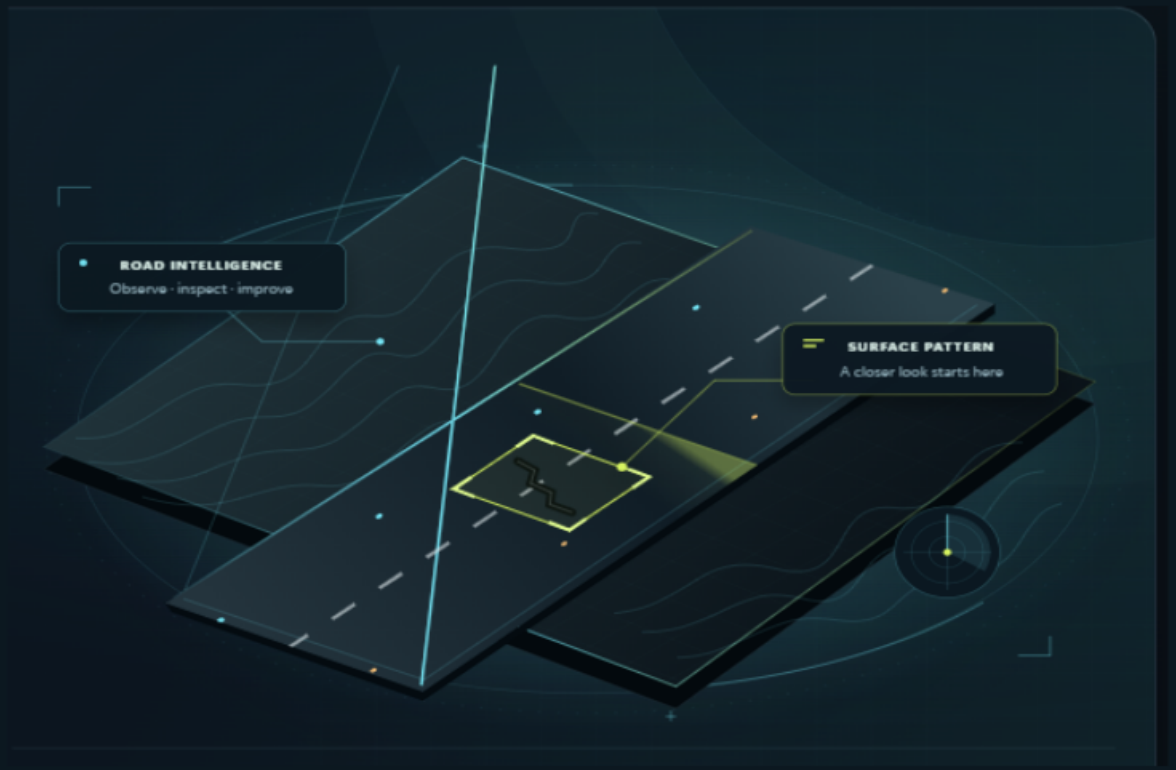
RoadScan

Human-reviewed road intelligence

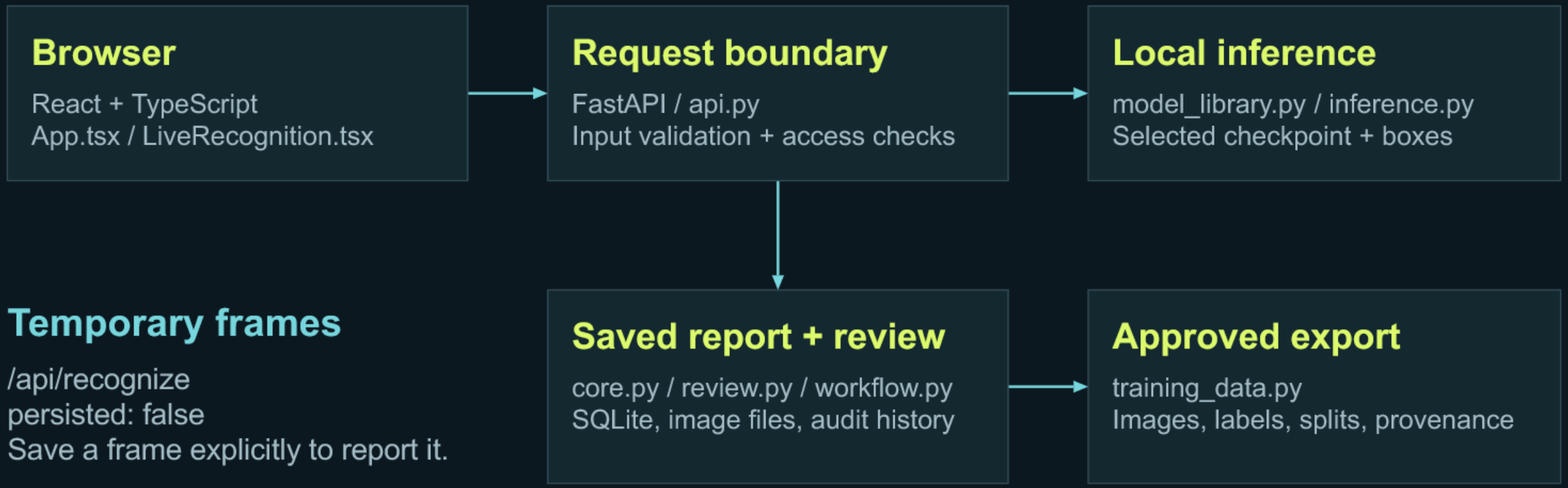
A local workflow for detecting damage, reviewing evidence and preparing better training examples.



Original app illustration, not a model prediction

Feras Alyamani / Ahmed Salamah / Ahmed Alhejaili

How the code connects



Model prediction stays separate from the reviewer's final decision.

A person makes the final call

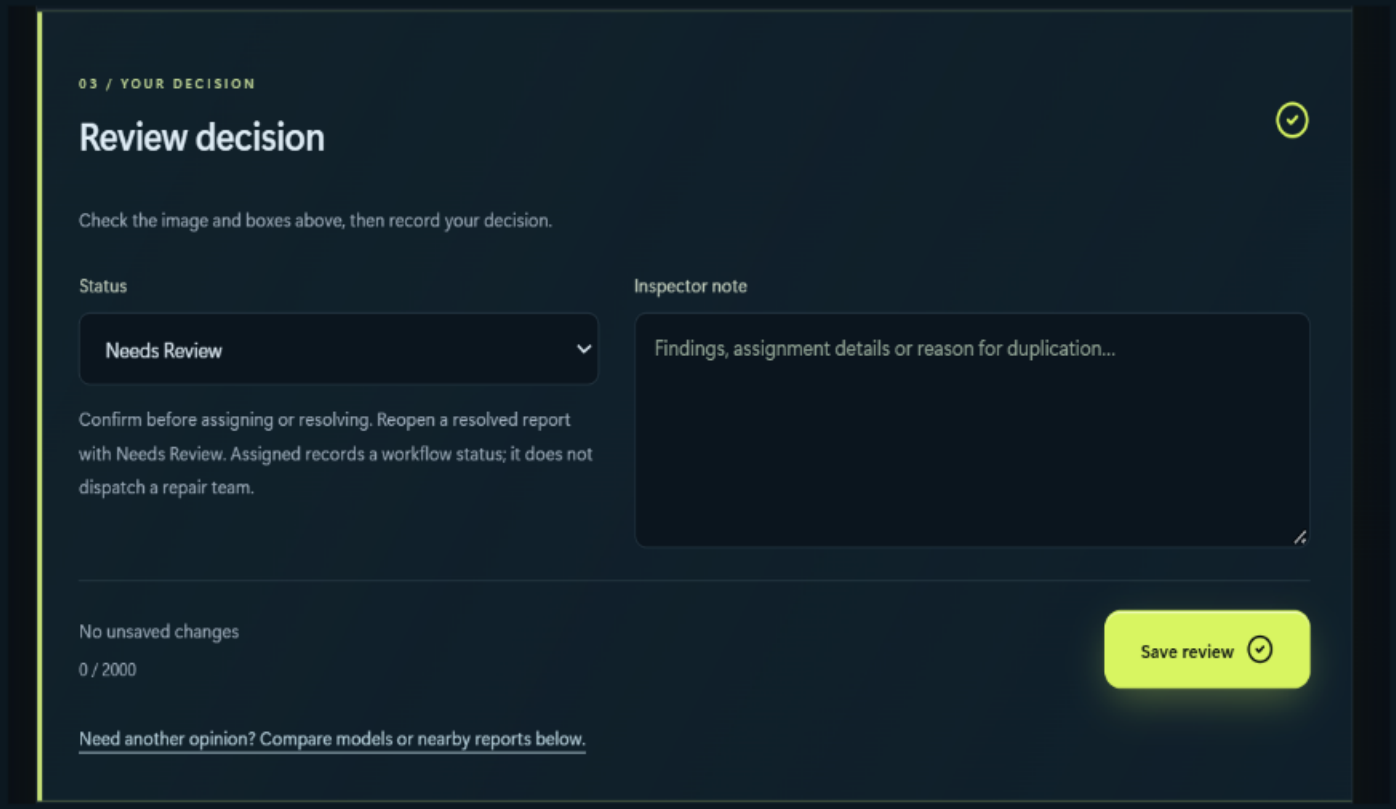
01 Inspect the boxes

Correct the class and location.
Keep the original prediction in history.

02 Compare model opinions

Choose any completed model,
including a minority result.
Its boxes become an editable draft.

03 Save the review



Priority supports triage. It does not measure physical crack depth.

Model results depend on the test set

Checkpoint	Training images	Evaluation scope	mAP50	mAP50–95
RoadScan baseline	8,000	US holdout 1,000 images	25.05%	10.96%
YOLO26s main	~41,000 reported	Team RDD test 5,758 reported	63.04%	33.01%
RT-DETR-L	Count unverified	Saved validation different split	61.77%	31.28%

Baseline: 8,000 train + 1,000 Czech validation + 1,000 US test

These rows are separate experiments, not a common-test leaderboard.

Team data includes reported near-duplicate leakage. Saudi field accuracy is still unmeasured.

Live viewing and complete video analysis

Camera / Iriun / USB

Current camera frame
Browser captures a fresh frame

→

Local detector
One request at a time

→

Result overlay
Responsive sampled recognition

Uploaded file: “Analyze every frame”

Decode next frame
FullVideoAnalysis.tsx

→

Wait for inference
video_analysis.py

→

Continue sequentially
Busy retry retains the same frame

Complete coverage can run slower than playback. A heavy model reduces throughput.

THE LEARNING WORKFLOW

How human review improves later models

Saving a review does not teach the model instantly. RoadScan collects approved examples; training happens separately.

01
Correct & save
In Review queue, fix the boxes and save as Confirmed.

02
Approve
On that report, choose Approve this example.

03
Export
Use the download button below. Share the ZIP with your model trainer.

04
Train separately
Train a new model in Colab or another training tool.

05
Test & import
Check it on unused test images, then import it in Settings.

No approved examples yet. Start with one report.

Review queue → correct boxes → save Confirmed → Approve this example.

Saved corrections leave the active model unchanged.

Approved export / separate training / held-out test / deliberate import

A working system with clear next tests

221
backend tests

24
frontend tests

67
browser checks

Recorded release checks. One optional backend test skipped.

Next evidence

Labeled Saudi road images and real repeat-damage pairs.
A shared model test and representative camera latency.

Code, reports and model downloads: github.com/Feras-MQ/RoadScan

The current 22-file model inventory

Role	Count	Purpose
Road detectors	19	Boxes and road-damage classes
ResNet18 classifier	1	Image-level classification
Stock YOLO initializer	1	General COCO initialization
DINOv2 embedding model	1	Optional visual similarity

Downloads: 11 road detectors + DINOv2. Nine files remain on hold.
The stock initializer links to its official upstream release.

Two safeguards in the actual code

Temporary recognition

roadscan/api.py:139–148

```
raw=await read_bounded(request,2*1024*1024)
image=decode_image(raw,max_pixels=2_000_000)

# Response excerpt
'persisted':False

finally:
    image.close()
```

A model vote stays a draft

frontend/src/App.tsx:337

```
setBoxes(value);
setEdited(true);
```

The interface asks the reviewer to check every box and save.

Abbreviated source excerpts. Validation and review history remain in their shared modules.

Running RoadScan on another laptop

1 **Docker Desktop**

Use Linux containers and the repository setup guide.

2 **Local settings**

Generate the local token and keep the data volume.

3 **Trusted checkpoint**

Import a compatible road model and confirm Active.

4 **First report**

Analyze a photo, inspect the boxes, then save a review.

Fresh-machine acceptance, Mac/ARM and physical drone integration remain unverified.

A short demonstration with evidence

Show

A road image and its prediction.
A corrected report and saved decision.
An individual model chosen from a vote.
The approved-example export workflow.

Explain

Confidence is not accuracy.
Priority is an estimated triage aid.
Training happens separately.
Location and drone support need representative field validation.

If hardware is unavailable, use the cleared interface screenshots and recorded evidence.

Experiments and remaining evidence

E1 Classifier and detector

Different tasks and metrics. Original run reproducibility is incomplete.

E2 Size and resolution

Reported accuracy trade-offs. Comparable CPU latency is still needed.

E3 Augmentation

Saved ablations include no augmentation and vertical flips.

E4 Country and road adaptation

Near-duplicate leakage limits claims. Saudi field testing is still needed.

E5 Repeated damage

Embedding simulations exist. Real repeat-damage pairs remain untested.